

Why they come back

Week 4, Lecture 1 · LangoBee to \$10K MRR

Summer 2026 · Operating a Solo Language App from Zero Revenue

What we'll cover

- The Hooked model: trigger, action, variable reward, investment
- External vs internal triggers: why order matters
- Streaks done honestly: what Duolingo learned
- The notification machine: CURR and channel protection
- LangoBee's three candidate hooks

The Hooked model

Four stages of a habit-forming product

Nir Eyal's Hook Model describes a cycle that, with enough repetitions, moves users from needing an external prompt to self-triggering on internal cues:

- **Trigger:** the cue that initiates behavior (external or internal)
- **Action:** the simplest behavior in anticipation of a reward
- **Variable reward:** the unpredictable payoff that keeps the user engaged
- **Investment:** the user adds data, effort, or stake that raises the cost of leaving

The Investment stage is what makes the next loop easier to complete. A streak, a saved word list, a progress record: all are investment.

(Eyal, 2021)

External triggers come before internal ones

An external trigger is a cue in the user's environment: a push notification, an email, a badge on an icon.

An internal trigger is an emotional state the user has associated with the product: boredom prompts opening Instagram; language anxiety prompts opening Duolingo.

The sequence matters. External triggers must repeat often enough that the user begins to self-trigger without them. A product that ships zero external triggers is waiting for users to form habits they were never given the chance to build.

LangoBee has sent zero push notifications. The internal trigger has never been seeded.

(Eyal, 2021)

Variable reward in a language app

Three types of variable reward (Eyal, 2021):

- **Tribe:** social validation (leaderboards, likes, shared progress)
- **Hunt:** searching for information or resources (a word you didn't know, a video that matches your level)
- **Self:** intrinsic satisfaction, mastery, completion (a text finished at 80% comprehension)

For LangoBee, the hunt reward is already live: each word tap surfaces a definition the user didn't know. The word_lookup_opened event fires 5,225 times per 30 days. That is the variable reward loop already running. The problem is nothing triggers the user to return to collect it again.

Investment: what LangoBee already has

The Investment stage creates stored value that raises the switching cost. LangoBee already accumulates:

- Word status records (known/learning) in the dictionary table
- Streak days in `courses.streak_days`
- Reading progress in `reading_progress`
- Comprehension percentages per text

None of these are currently surfaced at session end. The user has invested; the product has not told them so. Showing accumulated investment at the right moment is the streak surface hook.

(Eyal, 2021)

Duolingo's notification machine

The CURR finding

Jorge Mazal's team at Duolingo ran a growth model in 2018 that decomposed DAU into input metrics. The finding:

Current User Retention Rate (CURR) had 5x the impact on DAU of any other input metric.

New user acquisition, reactivation, and new user retention all mattered. But improving CURR for users who were already active compounded fastest because every current user kept was also a potential referrer and a potential payer.

The implication: before running a user acquisition campaign, fix CURR. This is why the retention phase precedes the distribution phase in this course.

(Mazal, 2023)

Three mechanics Duolingo shipped

After identifying CURR as the variable with the highest impact, Duolingo executed three interlocking mechanics:

1. **Streak optimization:** the streak became the product's primary investment mechanic. Loss aversion (fear of breaking a streak) proved more motivating than achievement (pride in building one).
2. **Gamified leaderboard leagues:** social pressure from a weekly league created a recurring external trigger to return before the week ended.
3. **Notification channel protection:** Duolingo treated each notification channel as a limited resource. Sending too often degraded open rates for everyone. They capped sends per user per day.

(Mazal, 2023)

The protect-the-channel rule

"One of the most important things we learned was to treat each notification channel as finite. Every send that doesn't get opened trains users to ignore future sends."

This rule has direct implications for LangoBee at launch:

- The first push notification you send sets the open-rate baseline. Send it only when the content is genuinely relevant.
- An indiscriminate first send to all users produces low opens and high unsubscribes. You cannot un-ring that bell.
- At LangoBee's current volume (roughly 4 real WAU), each push send reaches a small enough audience that one bad send is recoverable. That is a window to experiment that larger products do not have.

(Mazal, 2023)

"We immediately saw that CURR had a gigantic impact on DAU -- 5 times the impact of the second-best metric."

Jorge Mazal, Chief Product Officer, Duolingo (2023)

How Duolingo Reignited User Growth, Lenny's Newsletter

LangoBee's three candidate hooks

Candidate 1: PWA push notification

What it is: a browser push sent when the user has not opened the app in 24 hours. Content: "New Spanish video at your level."

External trigger: yes (device-level notification)

Internal trigger it aims to cultivate: habitual check-in when learning urge arises

Implementation cost: medium (service worker, permission prompt, Edge Function send job)

Honest constraint: web push permission conversion is 5-15% in cold contexts. At current volume, this reaches roughly 3 users per send.

The permission prompt is a one-time irreversible ask. If the first send is low-quality, users block future notifications and that channel is permanently closed.

Candidate 2: Daily "new for you" email

What it is: a daily email surfacing one new Spanish short matched to the user's comprehension level, using the mean_zipf ranker and word status.

External trigger: yes (email inbox)

Internal trigger it aims to cultivate: checking email as a content-discovery habit

Implementation cost: medium (Edge Function, email template, Supabase relay)

Honest constraint: email open rates average 20-25%. LangoBee's user base is small enough that this is a qualitative, not quantitative, experiment. You are testing message quality, not statistical significance.

The email hook has the advantage of being undoable: unsubscribe is one click and does not block a future channel.

Candidate 3: Streak surface at session end

What it is: showing the user's current streak count and a "come back tomorrow" message at the end of every reading or viewing session.

External trigger: none (this is an internal-trigger amplifier only)

Internal trigger it aims to cultivate: loss aversion around streak continuity

Implementation cost: low (front-end only, schema already exists)

Honest constraint: streaks require two return visits to become motivating. At D1 below 30%, most users never see their streak above 1. The hook amplifies retention; it does not create it.

If you ship only one hook this sprint, the streak surface is the fastest to build and has no permission or deliverability risks.

Take-aways

1. The Hooked model has four stages: trigger, action, variable reward, investment. External triggers must precede internal ones because habits require repetition before they become self-sustaining.
2. Duolingo found that Current User Retention Rate had 5x the DAU impact of any other input. Fix CURR before spending on acquisition.
3. Notification channels are finite. Protecting the channel means sending only when the content is relevant, starting from the first send.
4. LangoBee's word-lookup loop is already a working variable reward cycle. The missing piece is the trigger that brings users back to collect it.
5. Three hooks are available: PWA push (medium cost, channel risk), daily email (medium cost, undoable), streak surface (low cost, internal-trigger only). Choose one and ship it this sprint.

What's next

- **Reading:** Week 4 reading at </c/langobee-10k-mrr-26su/readings/wk04>: Hooks, triggers, and retention mechanics
- **Lecture 2 (this week):** Instrumenting return: D1/D7/D30 definitions, cohort curve shapes, small-n honesty
- **Section this week:** Choose your hook, write its event spec, build the D1 dashboard
- **HW2 (in flight, due week 5):** Ship the chosen hook end to end